

Main Identity

From: [REDACTED]
To: [REDACTED]
Sent: Tuesday, March 18, 2003 12:43 PM
Subject: RE: NDA fax??

Ted,

My fax machine doesn't work anymore. However, things can be faxed to [REDACTED] at [REDACTED]. His fax is broken at just this moment, but he will put in a new machine tonight. So tomorrow should be okay. Or just mail it here to the house.

VERY CLOSE HOLD. PROPRIETARY. Sadly, we found Johnson in very bad physical condition indeed. The machine is not ready to go, and he does not have the robust demonstrator. His physical condition is now so feeble that even two hours in the lab completely exhausts him. Every afternoon he has to go to bed and onto a machine to help his breathing, just to survive. He is physically unable to finish it, gather together the necessary instrumentation, etc. And he still is working completely solo; he has no assistance at all. The one backer he had who was a faithful friend and supporter, has been just about destroyed financially (that's how to take out Johnson, then just wait for Howard to die. That is apparently what has been done).

So we left there in very deep sadness and profoundly shocked. I fear that may be the last time I ever see Howard alive. If we succeed in getting funding for our MEG (we are still in some promising negotiations), we can perhaps pull things out for Johnson, by hiring a young fellow or two to work directly with him. But Howard has just about completely reached the end of his rope. He has been my firm friend for many years, since the 1970s, and this has troubled me deeply. He is simply unable to finish his work himself, and it is only going to get worse, rather than better. Also, many of his early patents have now expired, and he only has a couple or so still applying. And these only have two or three years left. He also needs patenting assistance, to cover his newer claims still not covered, which we are going to try come up with the funds etc. to help (perhaps with our own patent attorney). E.g., I can personally write a PPA if necessary, and file it for Howard since that only costs \$75 and the effort.

At any rate, SEAS should not plan on a [REDACTED] demonstrator, for that is still a long ways off, sadly. **END OF CLOSE HOLD.**

The trip was very exhausting for me, so I also got a gauging of my own physical situation. It's a bit worse than I thought or hoped. I really cannot afford to travel like that, even in a great car with the guys doing all the driving etc., unless it's nearly a life or death matter. It's going to take me several days to recover, it seems.

In case something untoward transpires with me, here are some rough but startling and revolutionary things that come out of my review of [REDACTED] rigorous new group theoretic methods establishing hierarchies of symmetry. His work is going to be the next great revolution in physics, almost certainly, as profound as was the discovery of broken symmetry with Nobel Prize awarded in 1957.

1. A broken symmetry at one level now also **GENERATES** a new symmetry at the next higher level, as well as "infolding" all the lower symmetries and broken symmetries. The present group theoretic methods used since the prediction and discovery of broken symmetry in 1957 generates only the **LOWER** level symmetry. It does not generate a higher reordering and symmetry. Hence it wrongly excludes nature's fundamental negative entropy mechanism, the generation (from disorder at one level) of order at the next higher level. This totally destroys the present second law of thermodynamics, which of course has been falsified experimentally to quite a significant degree already. My solution to the source charge problem (which Leyton's work beautifully fits and **GENERATES**) also falsifies the second law totally, because it includes the negentropy.
2. This resolves the present greatest problem in all of thermodynamics: It's asymmetry. In other words, for a century it has been severely puzzling that, if entropy can only remain the same or increase in progressive interactions, then how in the dickens did the entropy ever get so low in the first place, to provide the initial order? In other words, if one puts it bluntly, the presently accepted second law of thermodynamics is an oxymoron, implicitly assuming its own contradiction has first occurred. Now for the very first time we have

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the answer, rigorously, with rigorous group theoretic methods establishing it.

3. This work by [REDACTED] also solves what is perhaps the greatest problem in modern physics today: the problem that, with the uncovering of new broken symmetries, puzzlingly there also do appear new symmetries at higher levels. AS Weinberg points out, this problem has been driving much of the particle physics work. There has previously been utterly no justification for that appearance of new higher symmetry (a priori a negative entropy operation of consuming positive entropy to form negative entropy), since negative entropy was prohibited except for short statistical fluctuations as "temporary little flukes". Now there is a rigorous, continuous process, guaranteed mathematically. So what is perhaps the greatest problem in particle physics is also solved by [REDACTED] work.
4. Note that [REDACTED] and [REDACTED] were startled to find that a NESS (nonequilibrium steady state) system theoretically could produce negative entropy, and this negative entropy could increase more negatively toward negative entropy as time passes. They were rather startled, and felt that no real physical system could exhibit such a Gibbs entropy. However, being careful scientists, they carefully pointed out that the problem remained for deterministic dissipative systems. I noticed that the source charge is indeed such a dissipative system (emits observable photons in all directions continuously), and the resulting macroscopic fields comprised of those emitted photons are deterministic as a function of radius, to any size level desired and for any time duration desired. Hence that destroys completely the present second law of thermodynamics, which I've already written the formal resolution of, by changing it so the new form is consistent with experiment and known violations of the present second law formulation. The new statement is also consistent with Leyton's effect of hierarchies of symmetries, and in fact Leyton's work can be taken to predict that revised form of the second law.
5. I also found a minor flaw in the first law. Presently the first law defines the change of magnitude of an external parameter as work, and that is false for electrodynamics. The change of MAGNITUDE of potential energy in a given form, is just regauging and is freely permitted by gauge freedom. It is only the change of FORM of the energy that constitutes work. So the present first law is flawed, and it too must be revised — as presently stated, it specifically excludes gauge freedom, which is contradicted by a vast part of physics, both theoretically and experimentally. So we corrected that statement of the first law also. Energy and work CANNOT be logically equated! Work is the change of form of energy, not energy itself. There should never be a balance equation equating work to energy, and the units should actually be different. Energy is in joules, but work should be in CHANGE OF FORM OF INITIAL JOULES. All thermodynamics is wrong and flawed by this error, endemic throughout its entire structure. In an energy balance equation, work terms should only appear so as to mark where and when the FORM of energy is changed. All balance should be done with energy terms only, without any of the work terms. E.g., when energy is dissipated in a load, it is true that energy form is changed and work is done. However, the work has nothing at all to do with the ENERGY BALANCE. When the work is done and the form of the input energy is thereby changed, one still has all the energy remaining in its "dissipated" form. THAT is the term — the remaining energy in different form — that must be used instead of the work done, in any energy balance equation.
6. We also did a neat little analysis on the old saw of forbidden perpetual motion, which usual prohibition is just stupid. Newton's first law requires perpetual motion, once an entity is placed in any state of motion (including rest or zero motion), until the moving entity is acted on by an external force to change it. So perpetual motion is not only not forbidden, it is REQUIRED for every object set in motion. Else all mechanics and physics is destroyed. What is logically wrong is that the usual statements then equate "perpetual motion" as being a system performing work continuously, with zero energy input. That such a latter "continuous working system with no energy input" is forbidden, is quite true. But that has nothing to do with perpetual motion, which requires no energy input and does not work at all. To equate the two statements is to commit a logical non sequitur. So the standard stupid statements dealing with "dirty old forbidden perpetual motion) advance a false premise, then a true statement, then equate the two and assume that the true statement proves the false one! That is the simplest logical error that could be made; it is inexplicable that more than a century of thermodynamicists and physicists have just accepted such nonsense and logical violation without simple logical analysis. A machine that perpetually does work, e.g., but also perpetually (continuously) receives the necessary energy (from whatever source, including from the active environment), is jolly well permitted; we use that principle everytime we power up an electric motor or electrical machine.
7. Leyton's hierarchies of symmetry now clearly show the procedure for electrodynamics, from start to finish — - from the virtual particle flux of the vacuum to the production of the fields and potentials to the conservation of energy in the universe. Starting with the present virtual particle flux of the vacuum and the source charge, one can directly show the prediction and experimental emergence of those internested hierarchies of successfully higher symmetries. In the latest work I've fiddle with on it, it covers the

waterfront from the virtual energy of the vacuum throughout all intervening levels, to how the energy turns into observable energy on the charge, is radiated from the charge (deterministic dissipation process) to have deterministic ordered macroscopic fields and potentials (external parameters of the source charge as a system), how these "orderings" can then be dissipated (new broken symmetry at that level) to generate useful work in engines and devices, and then — and this is shockingly new — the automatic generation at the next higher level (the entire observable universe) of a great new symmetry and ordering — and that is the conservation of energy law itself!

8. The beautiful, beautiful thing I realized is that Leyton's effect means that the electrical charge as a thermodynamic system **GENERATES** its own external parameters, negentropically! Negentropy is now not a dirty word, but that half of the thermodynamics process that the old guys left out, because the steam in a steam engine does not generate its own external parameters (the cylinder, boiler, etc. to bound it and establish ordering). In electrodynamics, the fundamental system — the source charges — does indeed **GENERATE** its own system external parameters — the EM fields, potentials, and joules of observable EM energy of the universe. And that does not previously explicitly appear in thermodynamics and physics anywhere, except for the odd and erroneous assumption in electrodynamics that all EM fields and potentials and their energy are somehow **CREATED FROM NOTHING AT ALL** by their associated source charges (the long-vexing source charge problem, which I fundamentally solved in 1999 and published in 2000, but now have found rigorous support and proof for, in Leyton's work.
9. The energy conservation law — that energy can neither be created nor destroyed, but it can be changed in form — should have shown us something long ago. Entropy, being in effect a kind of "lost energy", can be changed in form (else there can be no conservation of energy law at all). But the only other form is negative entropy! So one can change entropy into negative entropy, and negative entropy into positive entropy, and [REDACTED] work clearly shows and proves exactly how that works, and the exact mechanism by which it works.
10. This is the next great revolution in physics, equal to or even greater than the discovery of broken symmetry in 1957.
11. These points are very, very important. Please keep them; should anything happen to me, these principles must not be lost, but must endure.

Best wishes,

[REDACTED]

-----Original Message-----

From: Te [REDACTED]mailto:[REDACTED]
Sent: Tuesday, March 18, 2003 9:45 AM
To: [REDACTED]
Subject: NDA fax??

Good Morning [REDACTED]

We tried to fax the copy of the NDA to you but, your fax machine did not seem to take it. Do we need to call first, or shall we just put it in the mail today? Perhaps easier that way.

Thanks, [REDACTED]